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Grouping method for the approximate solution of a kinetic equation describing the evolution of point-defect clusters

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ABSTRACT

An approximate method for the numerical solution of a kinetic equation describing the nucleation, growth and coarsening of point-defect clusters or second-phase precipitates is of interest for many applications in solid-state physics. In the present paper the validity of a grouping method developed by Kiritani (1973, *J. phys. Soc. Japan*, **35**, 95) is examined. In this method, which is the only one proposed up to now, a group of equations is replaced by an 'averaged' equation. Significant disagreement between the group kinetic equation and the original kinetic equation is revealed and the reasons for this are specified. A new grouping method based on the original equation is proposed and application of the method for the solution of two problems is demonstrated by comparing the results of numerical and analytical calculations.

§ 1. INTRODUCTION

In many important cases, changes in the physical and mechanical properties of materials during thermal treatment or under irradiation occur because of the formation and growth of point defect clusters (e.g. voids and dislocation loops) and second-phase precipitates. In order to understand these changes quantitatively, it is important to have a method for describing the microstructural evolution of clusters covering all stages from their nucleation to coarsening.

If random spatial distribution of the defect clusters is assumed, the microstructural evolution is described in terms of their size distribution function (SDF) $f(x, t)$. The SDF evolution obeys kinetic equations describing growth as well as dissolution of the clusters due to reactions with mobile defects and their thermal emission. For

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practical purposes, it is necessary to consider clusters containing a large number of point defects or atoms and the kinetic equations can be solved by means of numerical methods only. Such calculations are time consuming and, especially for high irradiation doses or long ageing times, can be performed only if the number of equations is reduced significantly. Grouping procedure provides such a reduction. The main idea is to replace a group of equations with an 'averaged' equation. Such a procedure was proposed by Kiritani (1973) for describing the evolution of vacancy loops during ageing of quenched metals. No alternative methods have been proposed since, and the Kiritani (1973) method (K method) has been applied to the solution of various problems, including those pertaining to irradiation conditions.

Having recognized the importance of the grouping method for further progress in the understanding of different aspects of the microstructural evolution in materials under various conditions, several papers have been devoted to the analysis and the validity of this method. Koiwa (1974) was first to examine the K method by comparing numerical results with the results of analytical solution for a simple problem. Serious disagreement was found between the numerical and analytical results and, therefore, strong doubts were raised regarding the applicability of the method. Hayns (1976) discussed the same problem in studying interstitial loop evolution under irradiation. Comparison of the results obtained numerically using the original and grouped equations for relatively low irradiation doses was performed. By doing so, Hayns tried to overcome Koiwa's objection by optimizing the grouping parameters, namely the group widths. Only some empirical observations were made, however. In general we conclude that a full and consistent analysis of the validity of the K method has not been available up to now.

In the present work we analyse rigorously the validity of the K method. In §§ 2 and 3 we describe the model and formulate criteria for developing an approximate method. We show analytically in § 4 and numerically in § 6 that the original kinetic equation and that introduced by Kiritani are different and we specify the reasons for this. The analysis leads to the conclusion that the Kiritani equation is inadequate. In § 5, we propose a new method which is based on consideration of the equations for the SDF moments and which satisfies all the criteria formulated. Application of the method is demonstrated in § 6 for two problems for which analytical solutions are available. The conclusions are briefly described in § 7.

§ 2. BASIC EQUATIONS

2.1. *Model description*

We consider vacancy cluster formation in a pure metal under irradiation and during ageing with the following assumptions.

- (1) In the case of irradiation, the primary damage is produced in the form of Frenkel pairs, that is single vacancies and self-interstitial atoms (SIAs).
- (2) The point defects execute a three-dimensional random walk.
- (3) The nucleation of vacancy clusters proceeds via a homogeneous mechanism, that is due to 'vacancy + vacancy = divacancy' reactions.
- (4) The clusters of point defects are immobile.

These assumptions, however, are only used for simplification and do not lead to any restrictions in the applicability of the analysis and the method developed.

2.2. Kinetic equation

Within the framework described above, we can write the following equation for the SDF $f(x, t)$ of clusters, (x is the number of defects in a cluster) (Katz and Wiedersich 1971):

$$\frac{\partial f(x, t)}{\partial t} = J(x-1, t) - J(x, t), \quad x \geq 2, \quad (1)$$

where $J(x, t)$ is the flux of clusters in space of cluster size:

$$J(x, t) = P(x, t)f(x, t) - Q(x+1, t)f(x+1, t). \quad (2)$$

In equation (2) $P(x, t)$ is the rate of vacancy absorption by a cluster of size x and $Q(x, t)$ is the sum of corresponding rates of SIA absorption (in the case of irradiation) and vacancy emission from the cluster. These rates depend on the configuration of the cluster and diffusion properties of the mobile defects. For a three-dimensional vacancy aggregate (i.e. void) and three-dimensionally diffusing point defects they take the following form:

$$P(x, t) = wx^{1/3}D_vC_v(t), \quad (3)$$

$$Q(x, t) = wx^{1/3} \left[D_i C_i(t) + D_v \exp \left(-\frac{E_v^b(x)}{k_B T} \right) \right] = Q_i(x, t) + Q_v(x), \quad (4)$$

where $w = (48\pi^2/\Omega^2)^{1/3}$, $C(t)$ and D are the concentrations and diffusion coefficients respectively of vacancies (subscript v) and SIAs (subscript i), $E_v^b(x)$ is the binding energy of a vacancy with a cluster of vacancies of size x , k_B is the Boltzmann constant, T is the absolute temperature and Ω is the atomic volume. For two-dimensional clusters (i.e. dislocation loops), P and Q have the forms (3) and (4) with $w = (4\pi/\Omega b)^{1/2}$ (b is magnitude of the cluster Burgers vector) and $x^{1/2}$ instead of $x^{1/3}$.

2.3. Equations for point-defect concentrations

In order to complete the system of equations (1)–(4), we must add equations for the evolution of the point-defect concentrations C_v and C_i , which are given by (Golubov 1981)

$$\begin{aligned} \frac{dC_v(t)}{dt} = & G + G^{\text{th}} - \mu_R D_i C_i(t) C_v(t) - k_v^2 D_v C_v(t) + Q_i(2, t) f(2, t) \\ & - [P(1, t) f(1, t) - Q_v(2, t) f(2, t)] - \sum_{x=1}^{\infty} [P(x, t) f(x, t) \\ & - Q_v(x+1, t) f(x+1, t)], \end{aligned} \quad (5)$$

$$\frac{dC_i(t)}{dt} = G - \mu_R D_i C_i(t) C_v(t) - k_i^2 D_i C_i(t) - \sum_{x=1}^{\infty} Q_i(x+1, t) f(x+1, t) \quad (6)$$

where G is the point defect generation rate, G^{th} is the vacancy emission rate from all sinks except vacancy clusters, μ_R is the recombination constant, and k_v^2 and k_i^2 are the sink strengths (excluding that of vacancy clusters) for vacancies and SIAs respectively. Note that the terms containing P and Q in equations (5) and (6) are consistent with those in equation (1).

Under ageing conditions, G and C_i (and consequently Q_i) are zero and equation (5) for the vacancy concentration is reduced to

$$\frac{dC_v(t)}{dt} = G^{\text{th}} - k_v^2 D_v C_v(t) - J(1, t) - \sum_{x=1}^{\infty} J(x, t). \quad (7)$$

2.4. Initial and boundary conditions

The initial and boundary conditions are taken to have the following form:

$$f(x, t=0) = C_{v0} \delta(x-1), \quad (8)$$

$$C_v(t=0) = C_{v0}, \quad C_i(t=0) = 0, \quad (9)$$

$$f(x=1, t) = C_v(t), \quad f(x=\infty, t) = 0, \quad (10)$$

where C_{v0} is the initial vacancy concentration, which is equal to the thermal equilibrium vacancy concentration in the case of irradiation and to the concentration C_q of quenched-in vacancies in the case of ageing.

Equations (1)–(6) are a set of coupled nonlinear equations and, in the general case, can be solved using numerical methods only.

§ 3. CRITERIA FOR APPROXIMATE EQUATIONS

In order to examine the validity of the K method, some criteria should be formulated. Development of the criteria is a challenge problem itself since an approximate method cannot be expected to provide exact solutions for any problem. Main properties of the exact solution should be maintained, however, and below we describe properties of equation (1), which we require to be retained in the approximate equations.

3.1. Limiting case $x \gg 1$

For clusters of large sizes, $x \gg 1$, equation (1) can be presented in the form of the Fokker–Planck equation

$$\frac{\partial f(x, t)}{\partial t} = -\frac{\partial}{\partial x} [V(x, t)f(x, t)] + \frac{\partial^2}{\partial x^2} [D(x, t)f(x, t)], \quad (11)$$

$$V(x, t) = P(x, t) - Q(x, t), \quad (12)$$

$$D(x, t) = \frac{P(x, t) + Q(x, t)}{2}. \quad (13)$$

In equation (11), $V(x, t)$ governs the increase in the mean cluster size, while $D(x, t)$ is responsible for broadening of the SDF.

3.2. Conservation laws

It follows from equation (1) that the total number of clusters given by

$$N(t) = \sum_{x=2}^{\infty} f(x, t) \quad (14)$$

and the total number of vacancies in the clusters given by

$$S(t) = \sum_{x=2}^{\infty} xf(x, t) \quad (15)$$

are described by the following equations:

$$\frac{dN(t)}{dt} = J(1, t), \quad (16)$$

$$\frac{dS(t)}{dt} = J(1, t) + \sum_{x=2}^{\infty} J(x, t). \quad (17)$$

Equation (16) shows that, in the framework of the model considered, $N(t)$ changes only owing to 'vacancy + vacancy $\leftrightarrow$ divacancy' reactions. Equation (17) is the conservation law for the total number of vacancies accumulated in the crystal. This law can be demonstrated in a particular case of ageing of clusters in a system containing no other sinks (i.e. $k_v^2 = 0$). Indeed, it then follows from equations (7) and (17) that in this case the total number of vacancies remains constant (note that $G^{\text{th}} \propto k_v^2$; hence it is equal to zero in the case considered)

$$\frac{d}{dt} \left(C_v(t) + \sum_{x=2}^{\infty} xf(x, t) \right) = 0. \quad (18)$$

3.3. Criteria

We propose the following criteria for the validity of the approximate solution of the kinetic equation (1). The approximate equations

- (a) must be identical with the original equations in the limiting case of group width $\Delta x_i = 1$ for all i (the subscript i indicates the number of a group),
- (b) must satisfy conservation law (16) for the total number of clusters,
- (c) must satisfy conservation law (17) for the total number of defects accumulated in system and
- (d) must transform to the Fokker-Planck equation (11) in the limiting case when $x \gg 1$.

§4. ANALYSIS OF THE KIRITANI METHOD

4.1. The Kiritani method

In the paper by Kiritani (1973), the clustering of quenched-in vacancies in the absence of other sinks (i.e. $k_v^2 = 0$) was considered. This case is described by equations (1) and (7). The SDF of the clusters with size up to $x = 100$ was calculated using equation (1). Larger clusters were grouped with the group widths $\Delta x_i = x_i - x_{i-1}$, where the subscript i indicates the number of a group and $x_i = x_{i-1} + 0.05(x_{i-1} + 1)$ (the i th group contains clusters of the sizes from $x = x_{i-1} + 1$ to x_i). The reaction rates were assumed to be constant within each group, that is $P(x, t) = P_i$ and $Q(x, t) = Q_i$. The evolution of clusters thus grouped was treated in terms of an average cluster concentration in a group F_i , the equation for which was written in the form similar to equation (1):

$$\frac{\partial F_i}{\partial t} = \frac{1}{\Delta x_i} (J_{i-1} - J_i), \quad (19)$$

where

$$J_i = \frac{2\Delta x_i}{\Delta x_i + \Delta x_{i+1}} P_i F_i - \frac{2\Delta x_{i+1}}{\Delta x_i + \Delta x_{i+1}} Q_{i+1} F_{i+1}. \quad (20)$$

4.2. Examination of the Kiritani method

It can be shown that equation (19) satisfies criteria (a), (b) and (c) formulated above. Whether or not it is consistent with criterion (d) is discussed in the following.

4.2.1. Satisfaction of criterion (d)

As can be seen, in equation (20) the original reaction rates $P(x)$ and $Q(x)$ (equations (3) and (4) respectively)), are replaced by the effective rates P_i^{eff} and Q_i^{eff} :

$$P_i^{\text{eff}} = \frac{2\Delta x_i}{\Delta x_i + \Delta x_{i+1}} P_i, \quad (21)$$

$$Q_i^{\text{eff}} = \frac{2\Delta x_i}{\Delta x_i + \Delta x_{i-1}} Q_i. \quad (22)$$

In the limiting case of clusters of large sizes (i.e. $x \gg 1$), equation (19) can be presented in the form of the Fokker–Planck equation (11) with

$$V_i = P_i^{\text{eff}} - Q_i^{\text{eff}}. \quad (23)$$

$$D_i = \frac{P_i^{\text{eff}} + Q_i^{\text{eff}}}{2}. \quad (24)$$

As can be seen, equations (12) and (13) and equations (23) and (24) for V and D are different. Indeed, in the case when $\Delta x_{i-1} < \Delta x_i < \Delta x_{i+1}$, which is usually used (for example Kiritani (1973) and Hayns (1976)), according to equations (23) and (24), V is smaller, while D is larger than that according to equations (12) and (13). (These features of the K method have already been highlighted by Koiwa (1974) and Hayns (1976).) These differences are very important since, as has already been pointed out, V governs the increase in the mean cluster size, while D is responsible for the broadening of the SDF. Thus, criterion (d) is not satisfied in the K method.

It is worth pointing out some of the main consequences of this on the cluster evolution. For ageing conditions, for example, this leads to an artificial deformation of the SDF of clusters; that is, the SDF becomes broader than the actual function and the mean cluster size becomes smaller (this is demonstrated numerically in § 6). The same outcome applies, but is much more serious, for irradiation conditions. Indeed the void swelling process, for example, is usually characterized in terms of the so-called bias factor and, as usual, the main aim of calculations is to reveal the consequences of, say, reactions or processes involved in the resulting bias factor. In this context, an artificial decrease in the mean cluster size in the K method could be interpreted as a corresponding artificial change in the bias factor. Consequently, it is rather difficult, if not impossible, to distinguish between the real and artificial effects in the results calculated using the K method.

4.2.2. Why criterion (d) is not satisfied in the Kiritani method

It is useful to understand why criterion (d) is not satisfied. In the K method, two main assumptions were used.

- (i) The cluster densities $f(x)$ are all equal within a group.
- (ii) Their increments are all equal within a group.

Below we show that these assumptions disrupt the authenticity of the process and should not be applied. (It will become clear from the following that another assumption in the K method that the reaction rates P and Q are equal within a group is not essential for the problem considered.)

Summing equation (1) over the i th group, we obtain the equation for the total number of clusters accumulated in the group:

$$N_i = \sum_{k=1}^{\Delta x_i} f(x_{i-1} + k). \quad (25)$$

This has the following form:

$$\frac{\partial N_i}{\partial t} = J(x_{i-1}) - J(x_i). \quad (26)$$

(Here and in the following the time dependence in J , P , Q , etc., is dropped.) If cluster densities are equal within a group, $f(x) = \tilde{F}_i$, then $N_i = \tilde{F}_i \Delta x_i$. Substituting this into equation (26), we obtain

$$\frac{\partial \tilde{F}_i}{\partial t} = \frac{1}{\Delta x_i} [J(x_{i-1}) - J(x_i)]. \quad (27)$$

Equations (27) and (19) look identical but in fact are different. Indeed, the right-hand side of equation (27) for the clusters in the i th group depends on the width of this group only, whereas in the K method the dependence on widths of the two adjacent groups is included.

Equation (27) satisfies criterion (a). Summing equation (27) over all cluster sizes, it can be shown that the resulting equation satisfies equation (16) (i.e. criterion (b)). It does not satisfy, however, the conservation law (17) for the number of vacancies in the clusters (i.e. criterion (c)).

An equation that satisfies criterion (c) can be obtained by multiplying equation (1) by x and summing over the group. As a result, we obtain the equation for the total number of vacancies in clusters of this group:

$$S_i = \sum_{k=1}^{\Delta x_i} (x_{i-1} + k) f(x_{i-1} + k) \quad (28)$$

in the form similar to equation (17),

$$\frac{\partial S_i}{\partial t} = (x_{i-1} + 1)J(x_{i-1}) - x_i J(x_i) + \sum_{k=1}^{\Delta x_i - 1} J(x_{i-1} + k). \quad (29)$$

If cluster densities are equal within the group, $f(x) = \tilde{F}_i$, then $S_i = \tilde{F}_i \langle x \rangle_i \Delta x_i$, where $\langle x \rangle_i$ is the mean cluster size in the i th group. Substituting this into equation (29) and taking into account that $x_i = \langle x \rangle_i - \Delta x_i/2$ and $x_{i+1} = \langle x \rangle_i + \Delta x_i/2$, one finds that

$$\frac{\partial \tilde{F}_i}{\partial t} = \frac{1}{\Delta x_i} \left(J(x_{i-1}) - J(x_i) - \frac{\Delta x_i - 1}{2\langle x \rangle_i} [J(x_{i-1}) - 2\langle J \rangle_i^* + J(x_i)] \right), \quad (30)$$

where $\langle J \rangle_i^*$ is the mean flux inside the i th group:

$$\langle J \rangle_i^* = \frac{1}{\Delta x_i - 1} \sum_{k=1}^{\Delta x_i - 1} J(x_{i-1} + k). \quad (31)$$

Note that, in contrast with equation (19) and similarly to equation (27), the right-hand side of equation (30) for the clusters in the i th group depends on the width of this group only. Summing equation (30) over all cluster sizes, it can be shown that the resulting equation satisfies equation (17) (criterion (c)), but equation (30) does not satisfy the conservation law (16) for the total number of clusters (i.e. criterion (b)).

Thus, equations (27) and (30) do not satisfy simultaneously the conservation laws (16) and (17) (i.e. criteria (b) and (c)). The reason for this is the assumption (i) used that the number density of clusters of different sizes belonging to the same group are equal and, hence, each group of clusters is characterized by one number only. This conclusion is in agreement with that made by Koiwa (1974) that this assumption 'can never be realised for any model system representing nucleation and growth processes'. In general, if it is required to satisfy two conservation laws, each group of clusters should be characterized by two numbers at least. Consequently, at least two equations for each group would be required.

In the K method (equations (19) and (20)) both conservation laws were satisfied, however, which contradicts the statement made above. This allows us to conclude that Kiritani (1973) achieved this satisfaction by using an *ad hoc* procedure, namely an artificial dependence on the widths of three adjacent groups was included. (The way in which it was done has been described on p. 96 after equation (2) of the paper by Kiritani (1973).) Indeed, the analysis performed above shows that the equation for the clusters in a group should not depend on the width of adjunct groups.

It is shown below both analytically (§ 5) and numerically (§ 6) that, as a consequence of this artificial dependence in the K method, the calculation results become very sensitive to the group widths (see also figures 1 and 2 in the paper by Koiwa (1974)). In other words, this disrupts the authentic physics of the process described by equation (1). The same reason lies behind the discontinuity of the cluster concentration at the interface between small ungrouped clusters and the larger grouped clusters revealed by Koiwa (1974).

§ 5. NEW GROUPING METHOD

In the following, a method is proposed to overcome the problems existing in the K method. This requires, however, that we must reformulate the original set of kinetic equations.

5.1. Reformulation of the kinetic equation

Let us consider an i th group of clusters containing Δx_i different cluster sizes. The group holds the same number of independent moments of the SDF: $M_m^i = \sum_{k=1}^{\Delta x_i} (x_{i-1} + k)^m f(x_{i-1} + k)$, $m = 0, 1, \dots, \Delta x_i - 1$. Equation (1) can be replaced by equations for the SDF moments within each group. These equations are easily obtained by multiplying equation (1) by x^m and summing over the whole

group. The system of equations for the SDF moments in each group is equivalent to the initial set of equations. The SDF within the group may be described, for example, in the form of a power series such that

$$f(x) = \sum_{m=0}^{\Delta x_i - 1} L_m^i (x - \langle x \rangle_i)^m, \quad (32)$$

where the coefficients L_m^i can be expressed through M_m^i .

Such a description gives us a clue as to how an approximate method for the solution of kinetic equations may be developed. Indeed, the number of equations in a group may be reduced by keeping only several of the essential moments (and, hence, the same number of terms in equation (32)). The problem then is to find the least number of SDF moments to be retained, without significantly disturbing the process. For any number, criteria (a) and (d) formulated in § 3.2 will be satisfied. However, any algorithm that satisfies criteria (b) and (c), that is the conservation laws (16) and (17), should include the equations for the first two SDF moments. The simplest algorithm, which considers only these two moments, is described below.

5.2. Simplest algorithm

Let us keep only equations for the first two SDF moments, which are easily obtained from equation (1) and approximate the SDF by a linear function within a group of different sizes of the type

$$f(x) = L_0^i + L_1^i (x - \langle x \rangle_i). \quad (33)$$

The coefficients L_0^i and L_1^i can be expressed through the zeroth and the first SDF moments. Substituting equation (33) into equations (25) and (28) for the total number N_i of clusters, and the total number S_i of point defects accumulated in the group, one can obtain

$$N_i = L_0^i \Delta x_i, \quad (34)$$

$$S_i = (L_0^i \langle x \rangle_i + L_1^i \sigma_i^2) \Delta x_i, \quad (35)$$

where σ_i^2 is the dispersion of cluster size in the group and is given by

$$\sigma_i^2 = \langle x^2 \rangle_i - \langle x \rangle_i^2 = \frac{1}{\Delta x_i} \left[\sum_{i=x_{i-1}+1}^{x_i} i^2 - \left(\sum_{i=x_{i-1}+1}^{x_i} i \right)^2 \right]. \quad (36)$$

Note that the grouping procedure is usually applied for large clusters (i.e. $x \gg 1$) when it becomes more practical to replace, with adequate accuracy, the sums in equation (36) by corresponding integrals. In this case, equation (36) is reduced to $\sigma_i^2 = \Delta x_i^2 / 12$.

Differentiating equations (34) and (35) over time and substituting into equations (26) and (29), one can obtain the following equations for L_0^i and L_1^i :

$$\frac{\partial L_0^i}{\partial t} = \frac{1}{\Delta x_i} [J(x_{i-1}) - J(x_i)], \quad (37)$$

$$\frac{\partial L_1^i}{\partial t} = \frac{\Delta x_i - 1}{2\sigma_i^2} [J(x_{i-1}) - 2\langle J \rangle_i^* + J(x_i)]. \quad (38)$$

Equations (37) and (38) describe the evolution of the SDF with the group approximation.

5.3. Main features of the new method

Equations (37) and (38) represent the grouping method which satisfies all the criteria formulated in § 3.3, namely they satisfy the conservation laws (16) and (17) and are identical with equations (1) and (11) in the limiting cases when $\Delta x_i = 1$ and $x \gg 1$ respectively.

It should be emphasized here that the new method differs substantially from the K method. First, in the method proposed here there is no problem in describing the SDF in the region between the ungrouped and grouped clusters. Indeed, equations (37) and (38), which depend on Δx_i only, transform to equation (1) at $\Delta x_i = 1$, that is $f(x_i) = L_0^i$ and $L_1^i = 0$, in contrast with the K method, where the equation describing the interface number density of clusters has a special form (see, for example, equation (21) in the paper by Koiwa (1974)). Second, the requirement imposed by the K method on the rates $P(x)$ and $Q(x)$ to be constant within a group is not necessary in the new method. Finally, for the same group widths, the new method requires twice as many equations as the K method but, as is shown in the next section, it is rather insensitive to the group widths and hence, much broader groups can be used.

§ 6. RESULTS AND DISCUSSION

The calculations presented below are illustrative and were performed for the following two cases: firstly, clustering of quenched-in vacancies in the framework of the Koiwa (1974) model; secondly, evolution of voids under irradiation assuming their homogeneous nucleation and neglecting vacancy emission from voids. The choice was determined by the fact that both problems have analytical solutions, which allows verification of the validity of both grouping methods by comparison not only between each other but also with exact solutions. The parameters used in the calculations were close to that of copper and are listed in table 1.

Table 1. Parameters used in calculations (dpa, displacements per atom).

<i>Irradiation and ageing</i>	
Temperature T	523 K
Vacancy diffusion coefficient D_v	$2 \times 10^{-6} \exp(-0.7 \text{ eV}/k_B T) \text{ m}^2 \text{ s}^{-1}$
Atomic volume Ω	$12 \times 10^{-30} \text{ m}^3$
<i>Irradiation</i>	
Point defect generation rate G	$10^{-7} \text{ dpa s}^{-1}$
SIA diffusion coefficient D_i	$10^{-6} \exp(-0.11 \text{ eV}/k_B T) \text{ m}^2 \text{ s}^{-1}$
Thermal equilibrium vacancy concentration C_{v0}	$\exp(-1.28 \text{ eV}/k_B T) \text{ atom}^{-1}$
Concentration C_q of quenched-in vacancies	$10^{-8} \text{ atom}^{-1}$
Dislocation density ρ_d	10^{15} m^{-2}
Dislocation capture efficiency Z_v for vacancies	1.00
Dislocation capture efficiency Z_i for SIAs	1.04
<i>Ageing</i>	
Concentration C_q of quenched-in vacancies	$10^{-8} \text{ atom}^{-1}$

6.1. Clustering of quenched-in vacancies

In this case we have used the model suggested by Koiwa (1974) (in particular, $k_v^2 = 0$, and $G^{\text{th}} = 0$ and $E_v^b(x) = \infty$). The grouping parameters were taken to be the same as in paper by Koiwa (1974), namely the clusters with sizes up to $x = 30$ were ungrouped while larger clusters were grouped with equal group size $\Delta x_i = \Delta x$. The value Δx was varied from 3 to 9.

Figure 1 presents the SDF of vacancy clusters at the end of the process when no single vacancies are left in the system (see details in the paper by Koiwa (1974)). The solid curve represents the analytical solution obtained by Koiwa (1974), while full and open symbols correspond to the results of the K method and the new grouping method respectively. As can be seen from the figure, the SDF obtained by the K method depends strongly on group widths and deviates from the exact analytical solution. That is, with respect to the analytical solution, the position of the SDF maximum (i.e. the most probable cluster size) is shifted to lower cluster size and the SDF is wider (i.e. the dispersion is larger). This is a manifestation of the smaller V and larger D described by equations (23) and (24) with respect to those given by equations (12) and (13), the fact already discussed in § 4.2. Note that the deviation of the SDF from the exact solution is larger for larger values of Δx . In addition, the discontinuity in the SDF at the crossover between single and grouped equations ($x = 30$) is observed (also reported by Koiwa) and $f(x = 30)$ depends strongly on the Δx chosen (note that in figure 1, for $\Delta x = 9$, $f(x = 30)$ is out of scale).

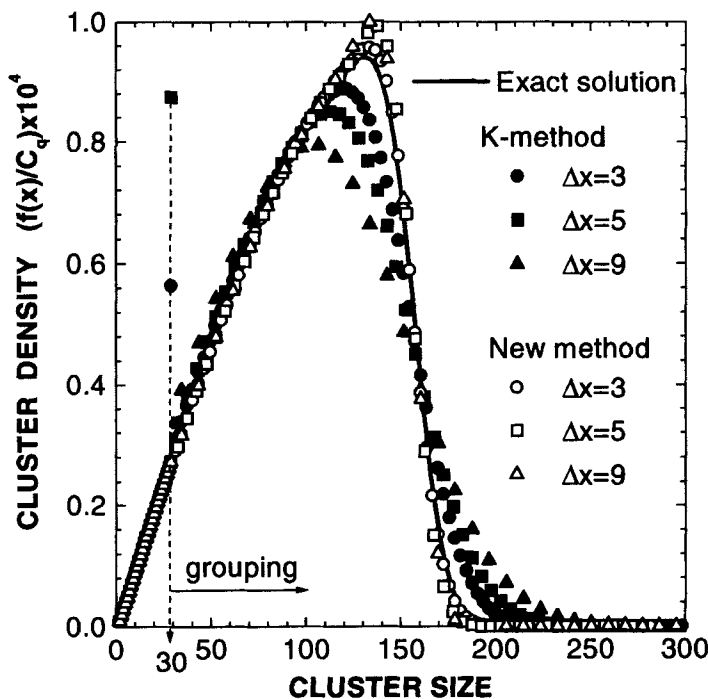


Figure 1. Terminal cluster SDF of vacancy clusters formed after ageing of quenched-in vacancies obtained analytically by Koiwa (1974) (—), using the K method ($\circ$, $\square$, $\triangle$) and the new method ($\bullet$, $\blacksquare$, $\blacktriangle$) for different group widths.

In contrast, the results obtained using the new method are in good agreement with the exact solution for all the group widths used. Some differences are observed only near the peak of the SDF. The reason for this is that the curvature of the SDF in this region is too large to be described by a linear approximation for the SDF within the groups. This shows a limitation of the method, which can be overcome either by decreasing the group widths or by increasing the number of SDF moments involved in the approximation of the SDF within the groups. However, in general, the calculation results obtained reproduce the exact solution quite well.

6.2. Nucleation and growth of voids under irradiation

To simplify comparison of the two grouping methods in this case, the model of void evolution was chosen, which is an irradiation analogy of the Koiwa (1974) model. Thus, it was assumed that voids are thermally stable (i.e. $E_v^b = \infty$ and, hence, $Q_v(x) = 0$), $G^{\text{th}} = 0$ and void nucleation occurs homogeneously. The point-defect recombination was neglected, and the sink strengths of dislocations for vacancies and SIAs ($n = v, i$), given by

$$k_n^2 = Z_n \rho, \quad (39)$$

(where ρ is the dislocation density, and Z_v and Z_i are the dislocation capture efficiencies for vacancies and SIAs respectively) were chosen to be much higher than that of voids for all doses. In this case, after a short transient period of irradiation, approximately $1/D_v \rho$, the concentrations of point defects reach the constant values

$$C_n \approx \frac{G}{D_n Z_n \rho}. \quad (40)$$

Under the conditions described above, the void nucleation rate $J_{\text{nucl}} \equiv J(1)$ remains constant during irradiation. This rate is obtained in the limit $Q_v(x) = 0$ from that derived by Katz and Wiedersich (1971):

$$J_{\text{nucl}} = w C_v (D_v C_v - D_i C_i). \quad (41)$$

This nucleation rate corresponds to a specific steady-state SDF $g(x)$ which satisfies the equation

$$J(x) = P(x)g(x) - Q(x+1)g(x+1) = J_{\text{nucl}} \quad (42)$$

and has the following form:

$$g(x) = \frac{C_v}{x^{1/3}}. \quad (43)$$

$g(x)$ can also be presented as a function of the void diameter d_v . Taking into account that $x\Omega = (\pi/6)d_v^3$, one can obtain

$$g(d_v) = g(x) \frac{dx}{dd_v} = \left(\frac{3\pi^2}{4\Omega^2} \right)^{1/3} C_v d_v. \quad (44)$$

The main prediction of the analytical analysis (Katz and Wiedersich 1971) is that the SDF $f(x, t)$ of voids, starting from the delta function (equation (8)), approaches $g(x)$. At any particular time, $f(x, t)$ is close to $g(x)$ in a certain range of void sizes, which increases with increasing dose. Below we use equations (41) and (44) to check the validity of the group methods for the case considered.

6.2.1. Calculations

The calculations were performed with non-uniform group widths $\Delta x_i = (x_i + 1)/M$ for $x > 30$ (also used by Kiritani (1973) and Hayns (1976)), where M is a variable. The evolution of SDF calculated is presented in figure 2. The broken curves correspond to the K method, and the solid curves to the method proposed here. As can be seen from the figure, the results obtained with these two methods differ significantly from each other. At all doses, the SDF obtained using the K method is wider, with the position of maximum shifted to lower size with respect to that obtained with the new method. This is a consequence of smaller $V(x)$ and larger $D(x)$ described by equations (23) and (24) with respect to those given by equations (12) and (13), the feature of the K method pointed out in § 4.2.

The function $g(d_v)$ is also shown in figure 2. As can be seen from the figure, the results obtained with the new method reproduce the prediction of the analytical analysis, namely that, at any particular irradiation dose, the SDF is close to $g(d_v)$ in a certain range of void sizes. In contrast, the results obtained using the K method differ significantly. Comparison of the void nucleation rate given by equation (41) leads to a similar conclusion.

Figures 3, 4 and 5 present the dose dependences of the void number density (calculated for voids with $x > 30$), the corresponding void sink strength and the value of swelling respectively for M equal to 10 and 30. Broken and chain curves correspond to the K method, and solid and dotted curves to the new method. As can be seen from the figures, the K method underestimates the void density, sink strength

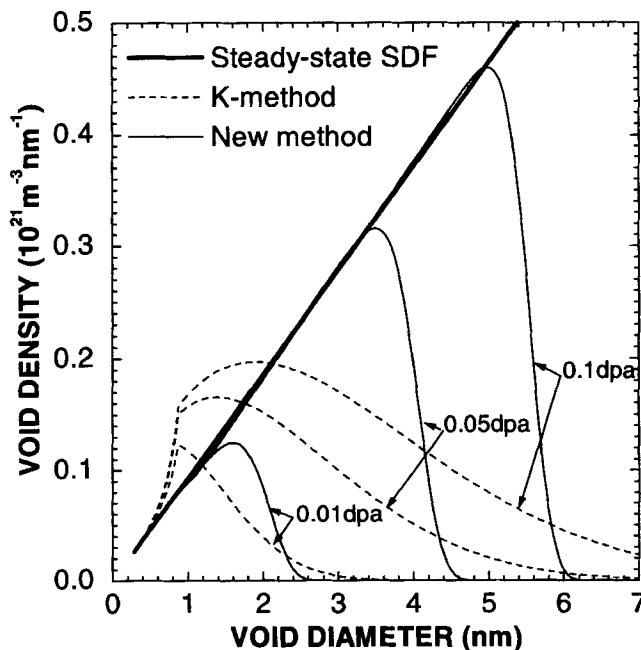


Figure 2. SDF of voids in copper irradiated at 523 K with the damage rate of 10^{-7} dpa s^{-1} for doses of 10^{-2} to 10^{-1} dpa. (---), the K method; (—), the new method; (—), the analytical function for the steady-state SDF (equation (44)). Note that the results calculated using the new method are consistent with the steady-state SDF, while those obtained by the K method differ significantly.

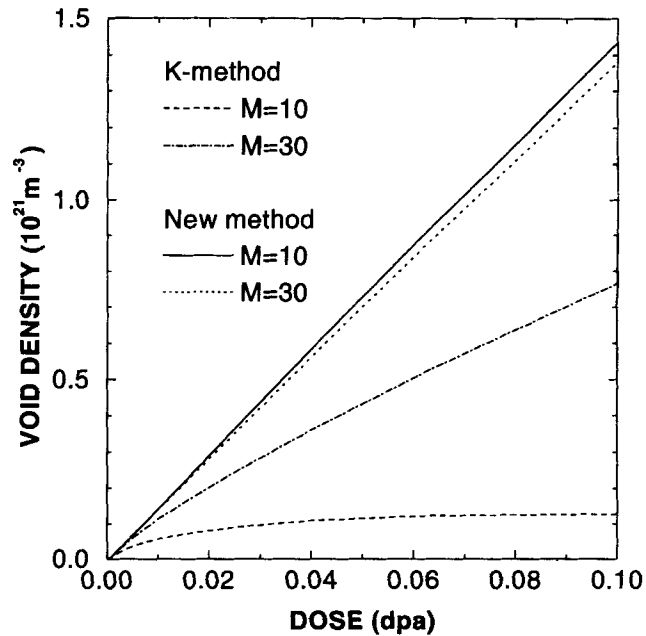


Figure 3. Dose dependence of the void density calculated by the K method (---, - · - · -) and the new method (—, ·····) for two different values of the parameter M . Note that the results calculated using the K method depend strongly on the choice of M , while those for the new method are almost independent of it.

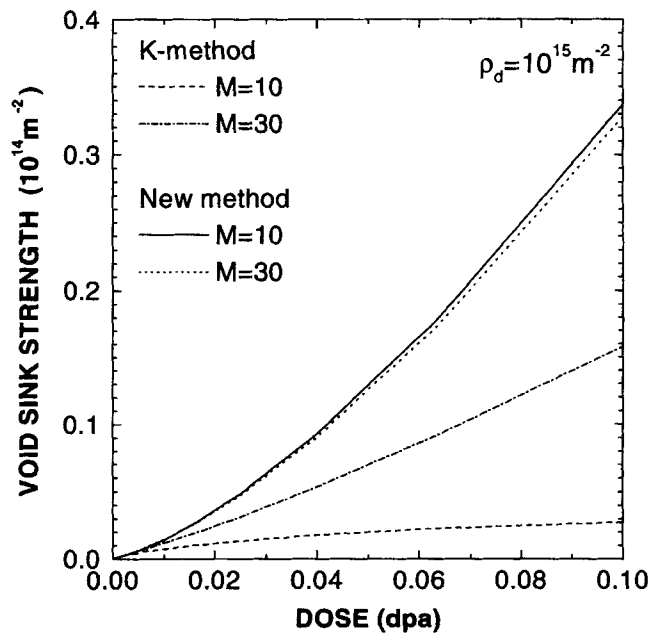


Figure 4. Dose dependence of the void sink strength calculated using the SDF given in figure 3. The notation is the same as in figure 3.

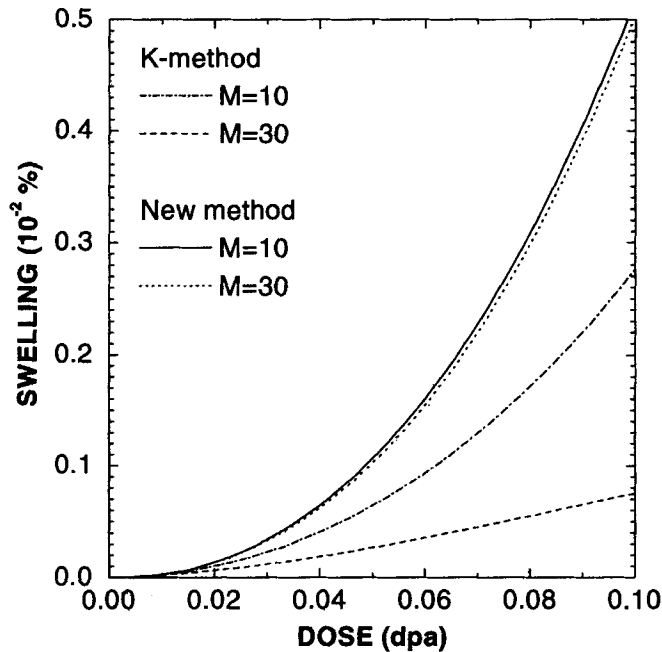


Figure 5. Dose dependence of the void swelling calculated using the SDF given in figure 3. The notation is the same as in figure 3.

and swelling level and the effect increases with increasing dose. In addition, the results obtained by the K method are shown to be very sensitive to the choice of M , whereas those obtained with the new method are almost independent of it.

There are some problems revealed in the application of the new method, however. That is, depending on the conditions and choice of the group widths, the SDF may show non-monotonic behaviour in such regions of cluster sizes where the size dependence of SDF within a group is too strong to be approximated by the linear function (equation (33)). If the groups are too wide, even negative values of the cluster concentrations could occur for the large cluster size (i.e. at the 'tail' of the SDF). As in § 6.1, we propose either to decrease the group widths or to increase the number of SDF moments involved in the approximation of the SDF within the groups to avoid this deficiency. It should be emphasized, however, that according to our calculations, this non-physical behaviour of SDF does not lead to any significant errors in the total number density of clusters and total number of point defects accumulated in the clusters. This demonstrates that the method proposed is rather stable in calculating integral values, that is the number density of clusters, swelling etc.

§7. CONCLUSIONS

- (1) Criteria for developing a grouping method for the approximate solution of the kinetic equation have been formulated, which are based on the properties of the initial equation that should be maintained in the approximate one.

- (2) It has been shown that the K (grouping) method does not satisfy the criteria and, hence, is inadequate. The reason for this is that the equation for the clusters in a group includes an artificial dependence on the widths of the adjacent groups and this leads to a distortion of the validity of the physical process described by the original kinetic equation.
- (3) A new grouping method, which is based on consideration of the SDF moments inside the groups and satisfies all the criteria, has been proposed. The method has been applied for calculation of the vacancy cluster evolution during ageing and under irradiation conditions. The results demonstrate good agreement with the analytical analysis.

The grouping method proposed allows generalization to the problems when the SDF is a function of two variables, for example gas-assisted nucleation when voids of equal size may contain a different number of gas atoms. In addition, it can be generalized to the case when one-dimensional diffusion of SIA clusters takes place together with the three-dimensional diffusion of point defects, which we have already performed and successfully used (Golubov *et al.* 1995, 2001, Singh *et al.* 1997).

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